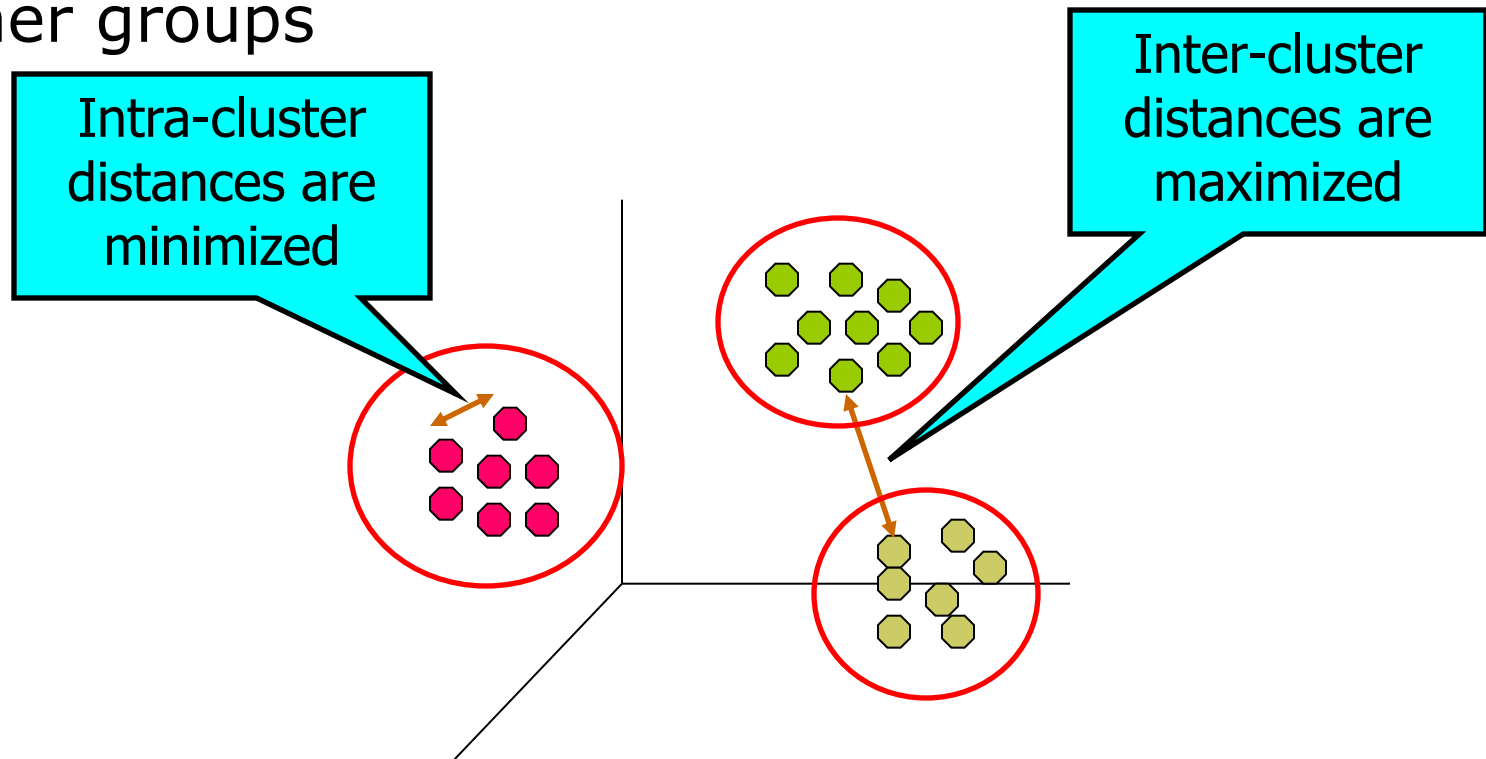


Cluster Analysis



What is Cluster Analysis?

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



What is Cluster Analysis?

- ❑ Cluster: a collection of data objects
 - Similar to one another within the same cluster
 - Dissimilar to the objects in other clusters
- ❑ Cluster analysis
 - Grouping a set of data objects into clusters
- ❑ Clustering is **unsupervised classification**: no predefined classes
- ❑ Clustering is used:
 - As a **stand-alone tool** to get insight into data distribution
 - ❑ Visualization of clusters may unveil important information
 - As a **preprocessing step** for other algorithms
 - ❑ Efficient indexing or compression often relies on clustering

Some Applications of Clustering

- Marketing: finding groups of customers with similar behavior given a large database of customer data containing their properties and past buying records;
- Biology: classification of plants and animals given their features;
- Libraries: book ordering;
- Insurance: identifying groups of motor insurance policy holders with a high average claim cost; identifying frauds;
- City-planning: identifying groups of houses according to their house type, value and geographical location;
- Earthquake studies: clustering observed earthquake epicenters to identify dangerous zones;
- WWW: document classification; clustering weblog data to discover groups of similar access patterns.

What Is Good Clustering?

- ❑ A good clustering method will produce high quality clusters with
 - high intra-class similarity
 - low inter-class similarity
- ❑ The quality of a clustering result depends on both the similarity measure used by the method and its implementation.
- ❑ The quality of a clustering method is also measured by its ability to discover some or all of the hidden patterns.

Requirements of Clustering in Data Mining

- ❑ Scalability
- ❑ Ability to deal with different types of attributes
- ❑ Discovery of clusters with arbitrary shape
- ❑ Minimal requirements for domain knowledge to determine input parameters
- ❑ Able to deal with noise and outliers
- ❑ Insensitive to order of input records
- ❑ High dimensionality
- ❑ Incorporation of user-specified constraints
- ❑ Interpretability and usability

Clustering Algorithms

Classification

Clustering algorithms may be classified as listed below:

- Exclusive Clustering
- Overlapping Clustering
- Hierarchical Clustering
- Probabilistic Clustering

Four of the most used clustering Algorithms

- K-means
- Fuzzy C-means
- Hierarchical clustering
- Mixture of Gaussians

Distances Measure

- An important component of a clustering algorithm is the distance measure between data points.
- If the components of the data instance vectors are all in the same physical units then it is possible that the simple Euclidean distance metric is sufficient to successfully group similar data instances.

K-Means Clustering Algorithm

K-means Clustering

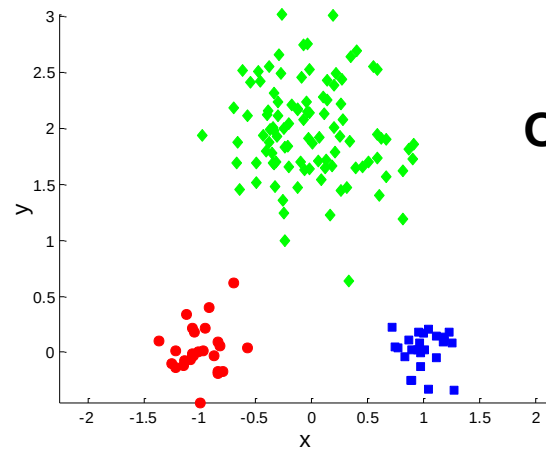
- ❑ Partitional clustering approach
- ❑ Each cluster is associated with a **centroid** (center point)
- ❑ Each point is assigned to the cluster with the closest centroid
- ❑ Number of clusters, K , must be specified
- ❑ The basic algorithm is very simple

-
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-

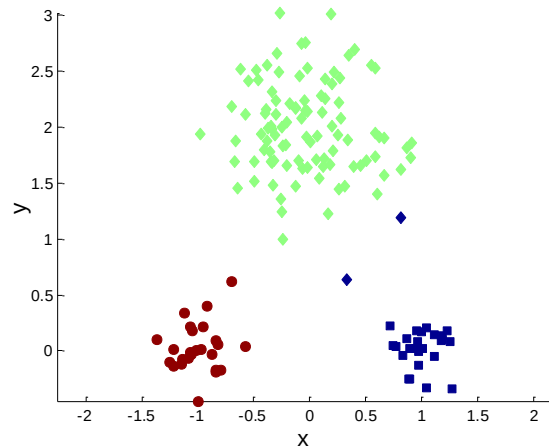
K-means Clustering – Details

- Initial centroids are often chosen randomly.
 - Clusters produced vary from one run to another.
- The centroid is (typically) the mean of the points in the cluster.
- 'Closeness' is measured by Euclidean distance, cosine similarity, correlation, etc.
- Most of the convergence happens in the first few iterations.
 - Often the stopping condition is changed to 'Until relatively few points change clusters'
- Complexity is $O(n * K * I * d)$
 - n = number of points, K = number of clusters, I = number of iterations, d = number of attributes

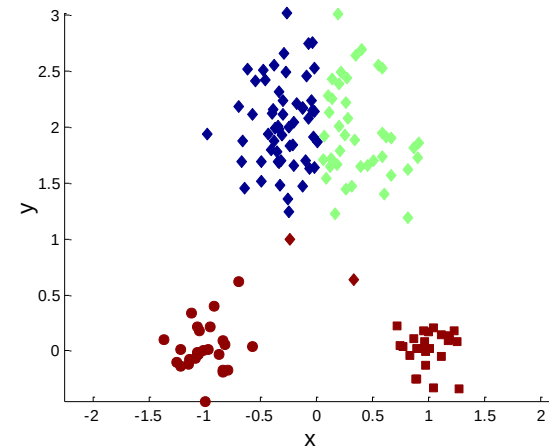
Two different K-means Clusterings



Original Points

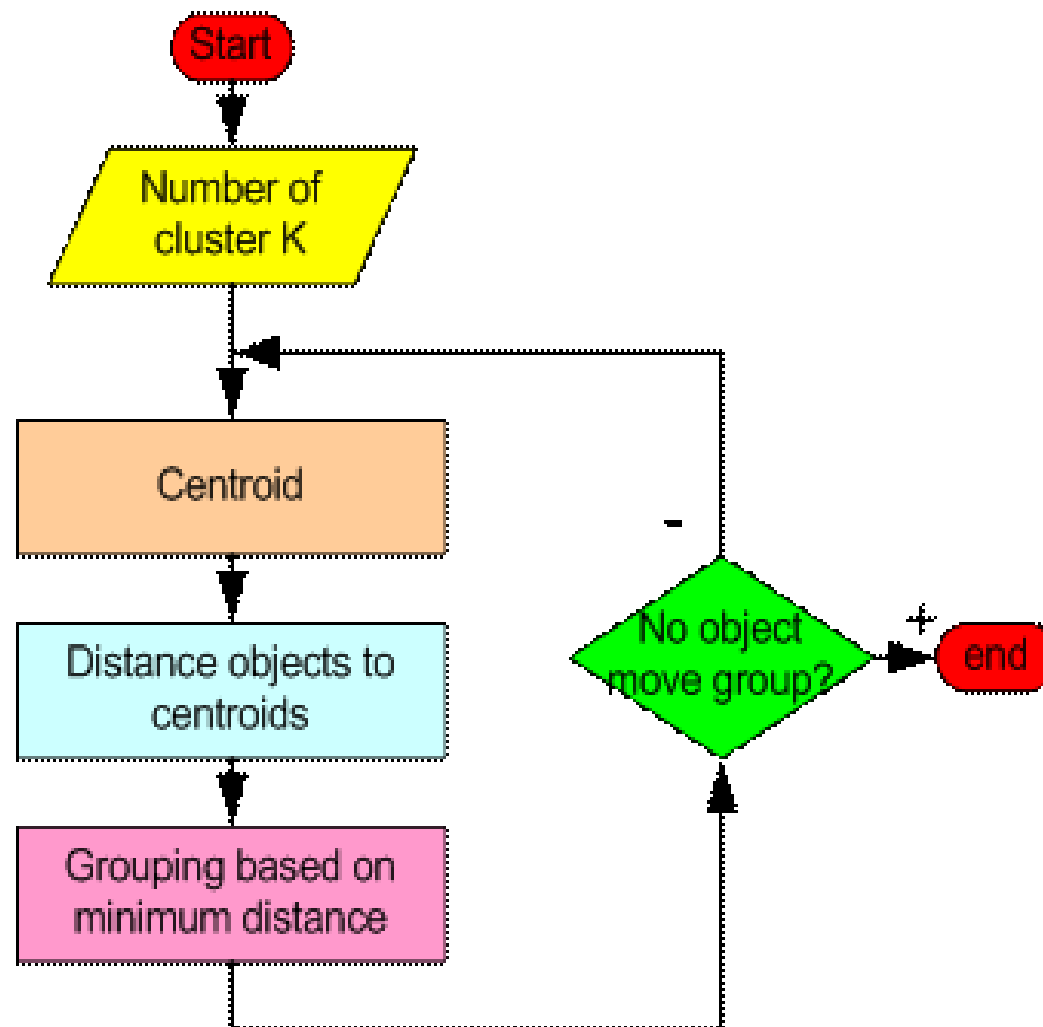


Optimal Clustering



Sub-optimal Clustering

How the K-Mean Clustering algorithm works?



Example of K-Means Clustering

We have 4 medicines as our training data points object and each medicine has 2 attributes. Each attribute represents coordinate of the object. We have to determine which medicines belong to cluster 1 and which medicines belong to the other cluster.

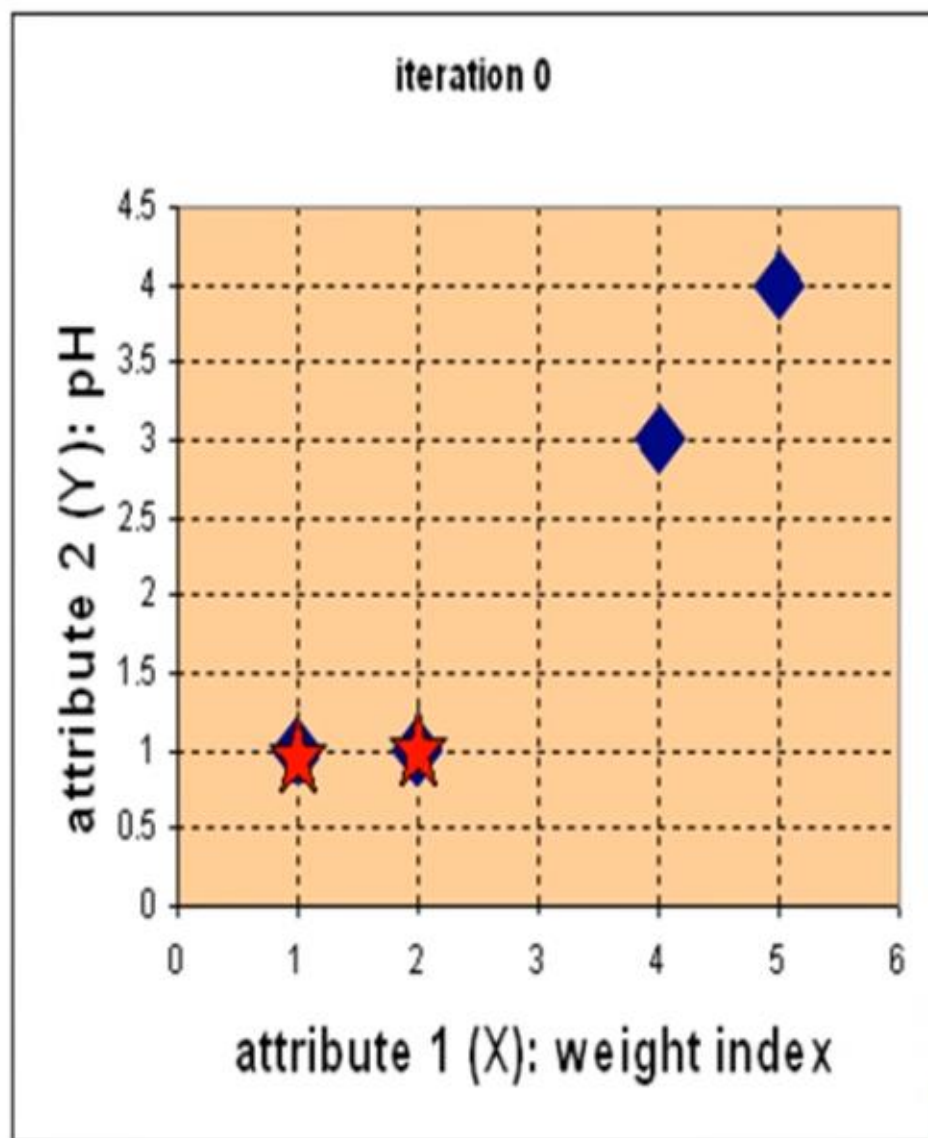
Object	Attribute1 (X): weight index	Attribute 2 (Y): pH
Medicine A	1	1
Medicine B	2	1
Medicine C	4	3
Medicine D	5	4

Step 1:

- **Initial value of centroids**

: Suppose we use medicine A and medicine B as the first centroids.

- Let c_1 and c_2 denote the coordinate of the centroids, then $c_1=(1,1)$ and $c_2=(2,1)$



- **Objects-Centroids distance** : we calculate the distance between cluster centroid to each object. Let us use Euclidean distance, then we have distance matrix at iteration 0 is

$$D^0 = \begin{bmatrix} 0 & 1 & 3.61 & 5 \\ 1 & 0 & 2.83 & 4.24 \end{bmatrix} \quad \begin{array}{l} \mathbf{c}_1 = (1,1) \text{ group-1} \\ \mathbf{c}_2 = (2,1) \text{ group-2} \end{array}$$

	A	B	C	D	
	1	2	4	5	X
	1	1	3	4	Y

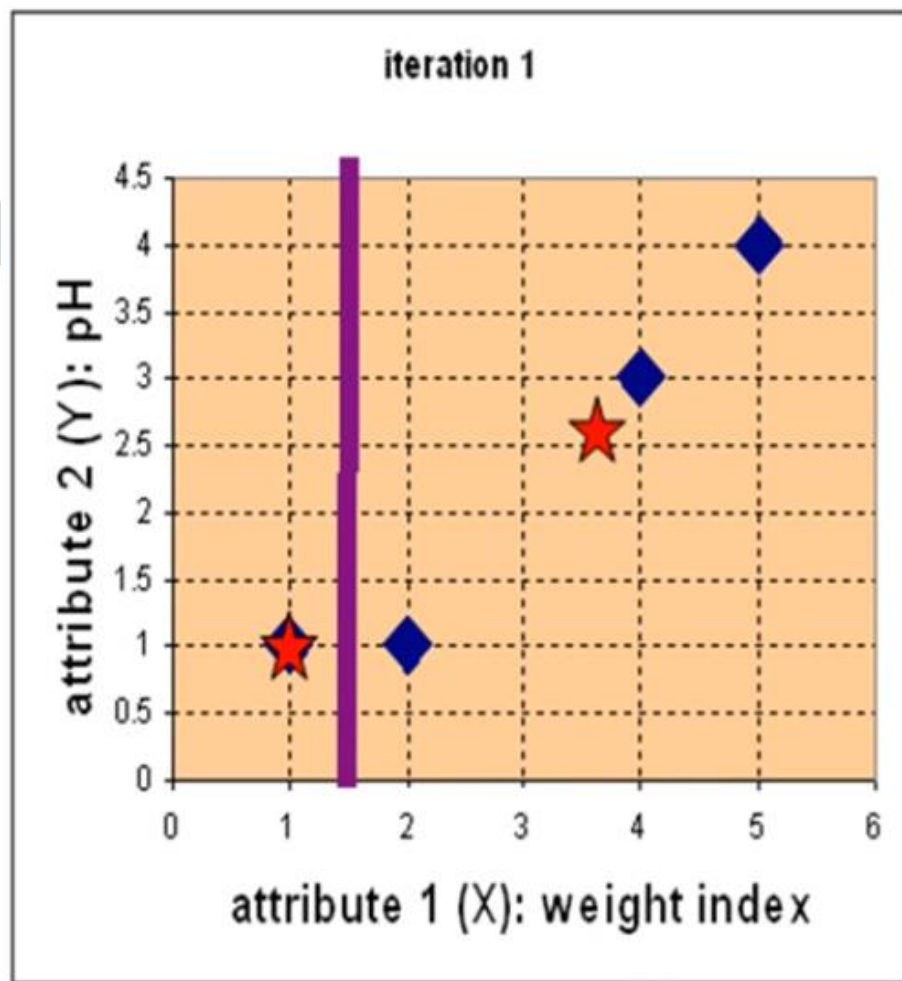
- Each column in the distance matrix symbolizes the object.
- The first row of the distance matrix corresponds to the distance of each object to the first centroid and the second row is the distance of each object to the second centroid.
- For example, distance from medicine C = (4, 3) to the first centroid $\mathbf{c}_1 = (1,1)$ is $\sqrt{(4-1)^2 + (3-1)^2} = 3.61$ and its distance to the second centroid is $\mathbf{c}_2 = (2,1)$ is $\sqrt{(4-2)^2 + (3-1)^2} = 2.83$ etc.

Step 2:

- **Objects clustering** : We assign each object based on the minimum distance.
- Medicine A is assigned to group 1, medicine B to group 2, medicine C to group 2 and medicine D to group 2.
- The elements of Group matrix below is 1 if and only if the object is assigned to that group.

$$\mathbf{G}^0 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{matrix} \text{group - 1} \\ \text{group - 2} \end{matrix}$$

A B C D



- **Iteration-1, Objects-Centroids distances** : The next step is to compute the distance of all objects to the new centroids.
- Similar to step 2, we have distance matrix at iteration 1 is

$$D^1 = \begin{bmatrix} 0 & 1 & 3.61 & 5 \\ 3.14 & 2.36 & 0.47 & 1.89 \end{bmatrix} \quad \begin{array}{l} \mathbf{c}_1 = (1,1) \text{ group-1} \\ \mathbf{c}_2 = (\frac{11}{3}, \frac{8}{3}) \text{ group-2} \end{array}$$

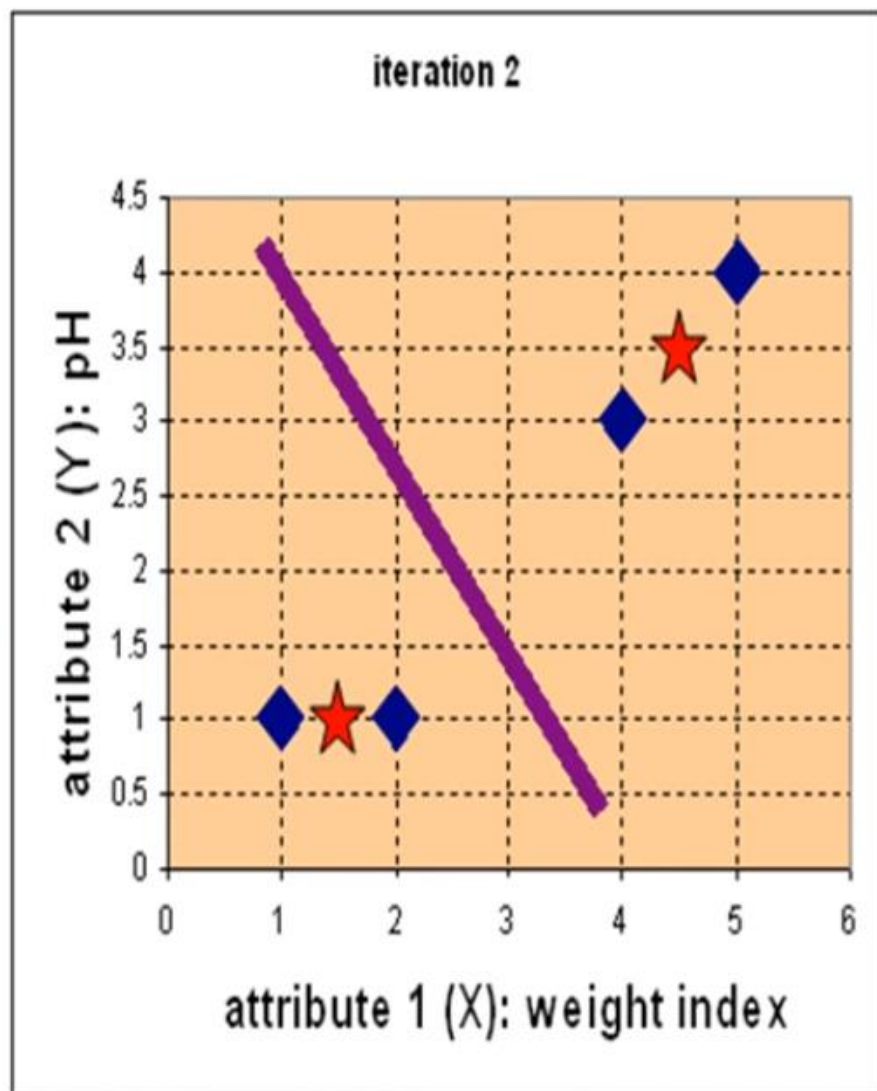
	A	B	C	D	
X	1	2	4	5	
Y	1	1	3	4	

- Iteration-1, Objects clustering:** Based on the new distance matrix, we move the medicine B to Group 1 while all the other objects remain. The Group matrix is shown below

$$G^1 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad \begin{matrix} \text{group - 1} \\ \text{group - 2} \end{matrix}$$

A B C D

- Iteration 2, determine centroids:** Now we repeat step 4 to calculate the new centroids coordinate based on the clustering of previous iteration. Group 1 and group 2 both has two members, thus the new centroids are
 $c_1 = (\frac{1+2}{2}, \frac{1+1}{2}) = (1\frac{1}{2}, 1)$
 and $c_2 = (\frac{4+5}{2}, \frac{3+4}{2}) = (4\frac{1}{2}, 3\frac{1}{2})$



- **Iteration-2, Objects clustering:** Again, we assign each object based on the minimum distance.

$$\mathbf{G}^2 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} \text{group - 1} \\ \text{group - 2} \end{matrix}$$

$A \quad B \quad C \quad D$

- We obtain result that $\mathbf{G}^2 = \mathbf{G}^1$ Comparing the grouping of last iteration and this iteration reveals that the objects does not move group anymore.
- Thus, the computation of the k-mean clustering has reached its stability and no more iteration is needed..

$G^i = G^{i+1}$ That the objects does not move group anymore

We get the final grouping as the results as:

<u>Object</u>	<u>Feature1(X): weight index</u>	<u>Feature2 (Y): pH</u>	<u>Group (result)</u>
Medicine A	1	1	1
Medicine B	2	1	1
Medicine C	4	3	2
Medicine D	5	4	2

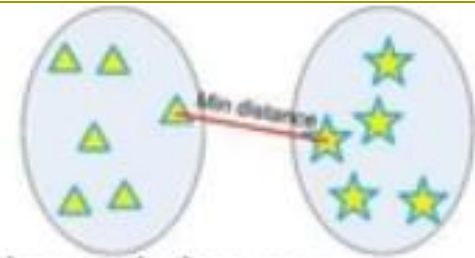
Hierarchical Clustering Algorithms

How They Work

Given a set of N items to be clustered, and an $N \times N$ distance (or similarity) matrix, the basic process of hierarchical clustering (defined by S.C. Johnson in 1967) is this:

- 1- Start by assigning each item to a cluster, so that if you have N items, you now have N clusters, each containing just one item. Let the distances (similarities) between the clusters the same as the distances (similarities) between the items they contain.
- 2- Find the closest (most similar) pair of clusters and merge them into a single cluster, so that now you have one cluster less.
- 3- Compute distances (similarities) between the new cluster and each of the old clusters.
- 4- Repeat steps 2 and 3 until all items are clustered into a single cluster of size N .

Step 3 can be done in different ways:



- **Single-linkage clustering** (also called the minimum distance method), we consider the distance between one cluster and another cluster to be equal to the shortest distance from any member of one cluster to any member of the other cluster.
- **Complete-linkage clustering** (also called the *diameter* or *maximum* method), we consider the distance between one cluster and another cluster to be equal to the greatest distance from any member of one cluster to any member of the other cluster.
- **Average-linkage clustering**

Example

Dist	A	B	C	D	E	F
A	0.00	0.71	5.66	3.61	4.24	3.20
B	0.71	0.00	4.95	2.92	3.54	2.50
C	5.66	4.95	0.00	2.24	1.41	2.50
D	3.61	2.92	2.24	0.00	1.00	0.50
E	4.24	3.54	1.41	1.00	0.00	1.12
F	3.20	2.50	2.50	0.50	1.12	0.00

We have 6 objects and we put each object into one cluster (analogue to put a ball into a basket). Instead of calling them as objects, now we call them clusters. Thus, in the beginning we have 6 clusters. Our goal is to group those 6 clusters such that at the end of the iterations, we will have only single cluster consists of the whole six original objects.

In each step of the iteration, we find the closest pair clusters. In this case, the closest cluster is between cluster F and D with shortest distance of 0.5. Thus, we group cluster D and F into cluster (D, F). Then we update the distance matrix (see distance matrix below). Distance between ungrouped clusters will not change from the original distance matrix.

How to calculate distance between newly grouped clustered (D,F) and other clusters?

- Using the input distance matrix, distance between cluster (D, F) and cluster A is computed as
 - $d_{(D,F) \rightarrow A} = \min(d_{DA}, d_{FA}) = \min(3.61, 3.20) = 3.20$
- Distance between cluster (D, F) and cluster B is
 - $d_{(D,F) \rightarrow B} = \min(d_{DB}, d_{FB}) = \min(2.92, 2.50) = 2.50$
- Similarly, distance between cluster (D, F) and cluster C is
 - $d_{(D,F) \rightarrow C} = \min(d_{DC}, d_{FC}) = \min(2.24, 2.50) = 2.24$
- Finally, distance between cluster E and cluster (D, F) is calculated as
 - $d_{E \rightarrow (D,F)} = \min(d_{ED}, d_{EF}) = \min(1.00, 1.12) = 1.00$
- Then, the updated distance matrix becomes

Min Distance (Single Linkage)

Dist	A	B	C	D, F	E
A	0.00	0.71	5.66	3.20	4.24
B	0.71	0.00	4.95	2.50	3.54
C	5.66	4.95	0.00	2.24	1.41
D, F	3.20	2.50	2.24	0.00	1.00
E	4.24	3.54	1.41	1.00	0.00

Looking at the lower triangular updated distance matrix, we found out that the closest distance between cluster B and cluster A is now 0.71. Thus, we group cluster A and cluster B into a single cluster name (A, B).

Dist	A,B	C	(D, F)	E
A,B	0	?	?	?
C	?	0	2.24	1.41
(D, F)	?	2.24	0	1.00
E	?	1.41	1.00	0

- Using the input distance matrix (size 6 by 6), distance between cluster C and cluster (D, F) is computed as

$$d_{C \rightarrow \{A,B\}} = \min \{d_{CA}, d_{CB}\} = \min \{5.66, 4.95\} = 4.95$$

- Distance between cluster (D, F) and cluster (A, B) is the minimum distance between all objects involves in the two clusters

$$d_{\{D,F\} \rightarrow \{A,B\}} = \min \{d_{DA}, d_{DB}, d_{FA}, d_{FB}\} = \min \{3.61, 2.92, 3.20, 2.50\} = 2.50$$

- Similarly, distance between cluster E and (A, B) is

$$d_{E \rightarrow \{A,B\}} = \min \{d_{EA}, d_{EB}\} = \min \{4.24, 3.54\} = 3.54$$

- Then the updated distance matrix is

Min Distance (Single Linkage)

Dist	A,B	C	(D, F)	E
A,B	0	4.95	2.50	3.54
C	4.95	0	2.24	1.41
(D, F)	2.50	2.24	0	1.00
E	3.54	1.41	1.00	0

Observing the lower triangular of the updated distance matrix, we can see that the closest distance between clusters happens between cluster E and (D, F) at distance 1.00. Thus, we cluster them together into cluster ((D, F), E). The updated distance matrix is given below.

Min Distance (Single Linkage)

Dist	(A,B)	C	(D, F), E
(A,B)	0.00	4.95	2.50
C	4.95	0.00	1.41
(D, F), E	2.50	1.41	0.00

Distance between cluster ((D, F), E) and cluster (A, B) is calculated as

$$d_{((D,F),E) \rightarrow (A,B)} = \min(d_{DA}, d_{DB}, d_{FA}, d_{FB}, d_{EA}, d_{EB}) = \min(3.61, 2.92, 3.20, 2.50, 4.24, 3.54) = 2.50$$

Distance between cluster ((D, F), E) and cluster C yields the minimum distance of 1.41. This distance is computed as

$$d_{((D,F),E) \rightarrow C} = \min(d_{DC}, d_{FC}, d_{EC}) = \min(2.24, 2.50, 1.41) = 1.41$$

After that, we merge cluster ((D, F), E) and cluster C into a new cluster name (((D, F), E), C).

The updated distance matrix is shown in the figure below

Min Distance (Single Linkage)

Dist	(A,B)	((D, F), E),C
(A,B)	0.00	2.50
((D, F), E),C	2.50	0.00

The minimum distance of 2.5 is the result of the following computation:

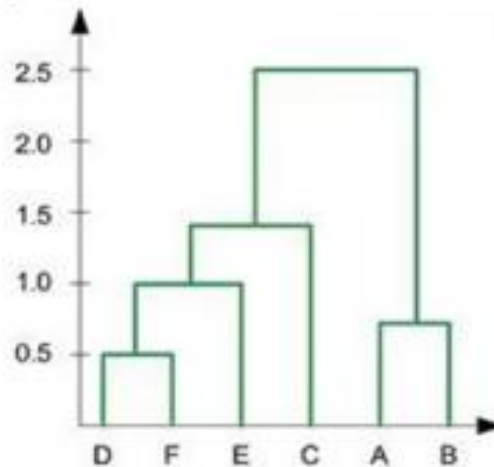
$$d_{(((D,F),E),C) \rightarrow (A,B)} = \min (d_{DA}, d_{DB}, d_{FA}, d_{FB}, d_{EA}, d_{EB}, d_{CA}, d_{CB})$$

$$d_{(((D,F),E),C) \rightarrow (A,B)} = \min (3.61, 2.92, 3.20, 2.50, 4.24, 3.54, 5.66, 4.95) = 2.50$$

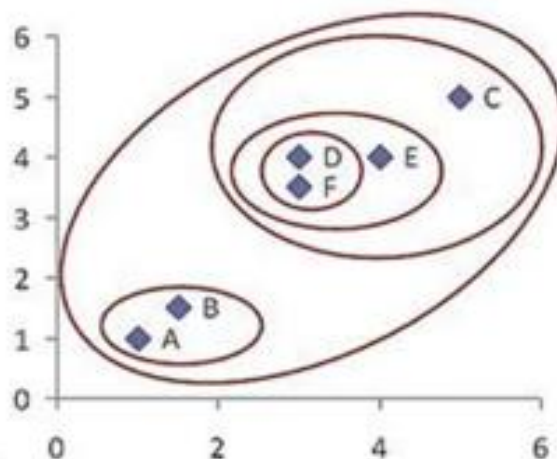
-
- Now if we merge the remaining two clusters, we will get only single cluster contain the whole 6 objects. Thus, our computation is finished. We summarized the results of computation as follow:

1. In the beginning we have 6 clusters: A, B, C, D, E and F
2. We merge cluster D and F into cluster (D, F) at distance 0.50
3. We merge cluster A and cluster B into (A, B) at distance 0.71
4. We merge cluster E and (D, F) into ((D, F), E) at distance 1.00
5. We merge cluster ((D, F), E) and C into (((D, F), E), C) at distance 1.41
6. We merge cluster (((D, F), E), C) and (A, B) into ((((D, F), E), C), (A, B)) at distance 2.50
7. The last cluster contain all the objects, thus conclude the computation

Using this information, we can now draw the final results of a dendrogram. The dendrogram is drawn based on the distances to merge the clusters above.



* The hierarchy is given as $((D, F), E), C, (A, B)$. We can also plot the clustering hierarchy into XY space



	X1	X2
A	1	1
B	1.5	1.5
C	5	5
D	3	4
E	4	4
F	3	3.5

Assignment

A hierarchical clustering of distances in kilometers between some Italian cities. The method used is single-linkage.

Input distance matrix ($L = 0$ for all the clusters):

	BA	FI	MI	NA	RM	TO
BA	0	662	877	255	412	996
FI	662	0	295	468	268	400
MI	877	295	0	754	564	138
NA	255	468	754	0	219	869
RM	412	268	564	219	0	669
TO	996	400	138	869	669	0



The process is summarized by the following hierarchical tree

